

Riana Moodley

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PERSONAL PROFILE

As an upcoming Electrical Engineering graduate, I am eagerly looking for opportunities to continuously keep learning new things as well as developing new skills.

EDUCATION

University of Zululand | *Beng Electrical Engineering*

Richards Bay | **02/2023 – Present**

- **Key Modules:**
 - Calculus
 - Physics for Engineers
 - Power Electronics and Machines
 - Power Systems

Hoerskool Richardsbaai | *Matric Certificate*

Richards Bay | **02/2018 – 11/2022**

UNIVERSITY/PERSONAL PROJECTS

Project 1: Swing Power Generator:

- Engineered a kinetic energy harvesting system that converts the mechanical motion of a swing into usable electrical energy.
- Explored energy conversion principles, generator design, and power conditioning circuits for a stable output.

Project 2: Smart Traffic Lighting System

- Designed an intelligent traffic management system that dynamically adjusts signal timing based on real-time traffic density.
- Sensor inputs and automated control logic work together to reduce congestion and improve intersection flow efficiency.

Project 3: Fertilizer Batch Mixing Automation:

- Developed a fully automated batch mixing system for fertilizer production using PLC-based control.
- The system precisely manages ingredient ratios, mixing sequences, and timing with industrial-grade logic for multi-stage process control.

SKILLS

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| • Skill 1: Proficiency in Microsoft Office Suite (Word, Excel, PowerPoint, Outlook) | • Skill 3: Effective time management. |
| | • Skill 4: Analytical thinking and problem solving |
| • Skill 2: Matlab, PSpice, PowerWorld Simulator, PSCAD knowledge. | • Skill 5: Adaptability to changing situations. |
| | • Skill 6: Detail orientation |

INTERESTS

- Taekwondo
- Running
- Gym
- Art